

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MASSIVE | Context: Ethiopian Intercity Travelers — Amharic-Language Booking Bot

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MASSIVE's Amharic content was created by 'tasking professional translators to localize the English-only SLURP dataset' [Q2, Q4], with seed data 'originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities' [Q106]. This means Amharic instances will not contain Ethiopian calendar dates (Meskerem, Tikimt — 13-month Ge'ez calendar offset 7–8 years from Gregorian, [WEB-4]), local city name variants (Adama/Nazret, Mekelle/Mek'ele), Birr-denominated fares, or holiday-surcharge contexts (Timkat, Meskel, Enkutatash — [WEB-5, WEB-6]) — all of which the user has confirmed are required slot values (Q2 elicitation). Worker-decided localization 'added further noise' [Q107], and per-language data is 'relatively small at 19.5k total records and 11.5k records for training' [Q103]. The IC dimension was rated HIGH priority and the mismatch is severe.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q103	"Starting with the dataset, the per-language data quantities are relatively small at 19.5k total records and 11.5k records for training." (p.9)
Q106	"Third, the data were originally created through crowd-sourcing, not from a real virtual assistant, which introduces artificialities." (p.9)
Q107	"Relatedly, allowing the worker to decide on translation versus localization of slot entities added further noise to the dataset, although we try to store this decision in the metadata." (p.9)
WEB-4	Ethiopian (Ge'ez) calendar: 13 months, 7–8-year offset from Gregorian — temporal expression slot values absent from MASSIVE (https://ethiocal.com/)
WEB-5	Timkat/Ethiopian Epiphany — major travel surge, Tir 11 E.C. (https://www.ethiopiancalendar.net/ethiopian-holidays)
WEB-6	Meskel, Genna, Enkutatash — peak travel periods unrepresented in MASSIVE (https://holidays.ertale.com/)

Strengths

- Translation workflow allows slot localization — workers could localize values like song names [Q57, Q58], so some Amharic content adaptation occurred
- Metadata stores the translate/localize/unchanged decision per slot [Q58], enabling researchers to filter for genuinely localized content

ID	Evidence
Q57	"The worker is asked to either localize or translate the slot, depending on whether the value should be translated (e.g., "tomorrow") or localized (e.g., the movie "La La Land", which in French is "Pour l'amour d'Hollywood.")" (p.6)
Q58	"The metadata of the released dataset includes whether the worker elected to "localize," "translate," or keep the slot "unchanged," primarily for the purposes of researchers evaluating machine translation systems, where it would be unreasonable to expect the system to "localize" to a specific song name the worker selected." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires Ethiopian calendar dates (Meskerem, Tikimt — [\[WEB-4\]](#)), local city names with Amharic spellings and alternate variants, Birr fare conventions, and operator/route knowledge specific to Sky Bus, Selam Bus, Ride. None of these are present in MASSIVE's English-derived content [\[Q2, Q4, Q106\]](#).

IC-2: MASSIVE's content originates from English crowd-workers [\[Q106\]](#) and was translated into Amharic [\[Q2\]](#); cultural alignment with Ethiopian intercity travelers is incidental. Holiday concepts, calendar conventions, and currency expressions are not Ethiopianized.

IC-3: Yes — the seed SLURP data reflects Western/Anglophone virtual-assistant interaction patterns (smart-home, music, weather domains [\[Q9, Q13, Q14\]](#)). Localization noise [\[Q107\]](#) further means inputs may carry residual Western reference frames.

IC-4: Recruitment of Ethiopian intercity-traveler-representative annotators is necessary; gap_id 5 confirms no L2-Amharic-speaker NLP dataset exists, and gap_id 11 confirms operator-specific data requires direct elicitation. INSUFFICIENT DOCUMENTATION on Amharic translator demographics in MASSIVE.

IC-5: Severe content-validity issues: absence of Ethiopian temporal expressions, locally varied city names, Birr fares, and code-switching with Oromo/Tigrinya (gaps 1, 2, 3, 4, 7) means the benchmark's Amharic inputs do not represent the target distribution. This is the dominant validity concern.

ID	Evidence
Q2	"MASSIVE was created by tasking professional translators to localize the English-only SLURP dataset into 50 typologically diverse languages from 29 genera." (p.1)
Q4	"MASSIVE was created by localizing the SLURP NLU dataset (created only in English) in a parallel manner." (p.1)
Q9	"The first iteration for the foundation of the MASSIVE dataset was the NLU Evaluation Benchmarking Dataset, with 25k utterances across 18 domains (Liu et al., 2019a)." (p.2)
Q13	"The Snips datasets (both the original English only and the English and French releases) are most similar to the NLU contained in the MASSIVE dataset, spanning smart home and music domains for a generic voice-based virtual assistant." (p.2)
Q14	"Facebook released a general Intelligent Virtual Assistant (IVA) dataset across the domains of Alarm, Reminder, and Weather (Schuster et al., 2019) created for the purpose of demonstrating cross-lingual transfer learning; and so did not need to be parallel or have an equal number of datapoints, resulting in far fewer examples in Thai (5k) compared to Spanish (7.6k) and English (43k)." (p.2)
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WEB-17	amseg Amharic segmenter/tokenizer (Yimam et al. 2021) exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)
WEB-21	No African transport-domain slot taxonomy exists — operator-specific slot types must be developed locally (https://aclanthology.org/2024.lrec-main.561/)

Information Gaps

- What fraction of Amharic slot instances in MASSIVE were marked 'localize' vs. 'translate' vs. 'unchanged' [Q58 metadata not summarized in paper]
- Whether any Amharic city names (e.g., Addis Ababa, Bahir Dar) appear in MASSIVE slot values

Requires Expert Verification

- Confirmation that Ethiopian intercity travelers' real utterances frequently mix Birr/Ethiopian-calendar/local-variant values with code-switching
 - Whether MASSIVE's Amharic data (despite limitations) provides sufficient typological prior for transfer learning to a finetuned booking model
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Remediation

Gap 1: Ethiopian-calendar dates, local city name variants (Adama/Nazret, Mekelle/Mek'ele), Birr fare expressions, and holiday-surcharge contexts are absent from translated MASSIVE Amharic data

Recommendation: Construct a supplementary slot-filling test set of 1–3k Amharic booking utterances elicited from Ethiopian intercity travelers, covering Ge'ez calendar months, alternate city names, Birr numerals (Arabic, Ethiopic, and verbalized forms), and named holidays (Timkat, Meskel, Enkutatash, Genna, Fasika, Eid). Use this as the primary acceptance test, with MASSIVE Amharic as a secondary baseline.